

Quantifying rates and drivers of change in long-term sector- and country-specific trends of carbon dioxide-equivalent greenhouse gas emissions

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ABSTRACT

Significance, direction, rate, and drivers of change in total carbon dioxide -equivalent ($\text{CO}_2\text{-eq.}$) greenhouse gas (GHG) ($\text{GHG}_{\text{CO}_2\text{-e}}$) emission from eight sectors of 41 Annex-I countries were quantified over the period of 1990–2012 using data-driven models. The Mann-Kendall (MK) test for trend analysis indicated an overall downward trend by 10.5% with a Sen's slope of $-613 \text{ Mt CO}_2\text{-eq./year}$. The highest $\text{GHG}_{\text{CO}_2\text{-e}}$ emission growth and reduction occurred with the sectors of energy, and land use, land use change and forestry (LULUCF) at rates of 1.2 and $-1.0 \text{ gigatons (Gt) CO}_2\text{-eq./year}$, respectively. Out of the 41 countries, 18 (44%) and 12 (29%) significantly reduced and increased their $\text{GHG}_{\text{CO}_2\text{-e}}$ emission rates, respectively ($p < 0.1$). The best-fit multiple non-linear regression (MNLR) model as a function of population density, gross domestic product per capita, fractional contribution of renewable energy to total energy production, country, and year elucidated 99.8% of variation in total $\text{GHG}_{\text{CO}_2\text{-e}}$ emission and had the predictive power of 99.8% based on leave-one-out cross-validation.

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1. Introduction

Svante Arrhenius (1896) is the first researcher to quantify and to speculate about impacts of the increased atmospheric concentration of carbon dioxide (CO_2) on the greenhouse effect and long-term variations in climate [1]. Charles D. Keeling performed the first continuous CO_2 measurement at 3400 m at Mauna Loa (HI, USA) in 1958 to better understand the relationship between

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CO₂ emitted from the burning of fossil fuels into the atmosphere and air temperature [20]. Since then, significant efforts have been made in the scientific arena to understand how the atmospheric concentrations of not only greenhouse gases (GHGs) including CO₂, water vapor (H₂O), methane (CH₄), ozone (O₃), nitrous oxide (N₂O), hydrofluorocarbon (HFC), chlorofluorocarbon (CFC), and perfluorinated carbon (PFC) but also aerosols influence the earth's energy budget and climate regime [2,8–10,12,17,24,36,40]. Mean global atmospheric CO₂ concentration has increased from the pre-industrial level of 277 ppm (ppm) to around 395 ppm currently, with an increase by 0.74 °C in global average air temperature since the pre-industrial times (1 Gt C=2.12 ppm) [17,26]. Mean annual global estimates between 2004 and 2013 were reported as 10.6 ± 2.9 and 9.5 ± 1.8 Gt CO₂ for terrestrial and oceanic sinks and as 32.7 ± 1.5 Gt CO₂ and 3.3 ± 1.8 Gt CO₂ for the human-induced sources of fossil fuel burning and cement production, and land-use change, respectively (1 Gt C=3.67 Gt CO₂) [4,26]. Globally, the total terrestrial and oceanic CO₂ sinks offset 43–69% of the total human-induced CO₂ sources [26].

Global climate change involves changes in air and soil temperature, daytime versus nighttime temperature, growing versus non-growing season temperature, precipitation regime, and extreme weather conditions [33,35,42,45]. These changes in turn have domino effects on snow and ice masses, sea level, timing and duration of seasonal activities of animals and plants, species distribution and abundances, spread of outbreaks and diseases, and socio-economic welfare [7,13–19,27–32]. Towards preventing and mitigating adverse impacts of GHG emissions on the well-being of humans and ecosystems, the Kyoto Protocol was signed internationally by 84 Parties in Kyoto (Japan) on 11 December 1997 so as to be committed to decrease GHG emissions by at least 5% of their 1990 levels over the five-year period (2008–2012), entered into force on 16 February 2005 and has been recently ratified by a total of 192 Parties [25]. Rogelj et al. [37] reported that total GHG emission should be reduced to the range of 41–47 Gt CO₂-equivalent (eq.)/year as the maximum appropriate target by 2020 so as to limit a rise in global average temperature to ≤ 2 °C above the pre-industrial levels called the threshold for the dangerous anthropogenic interference as the United Nations Framework Convention on Climate Change (UNFCCC) Parties agreed.

Our capacity to assist policy institutions and individuals in making better-informed decisions as well as to monitor international climate change cooperation and country-specific performances for any climate stabilization target depends on the development of data- and model-based assessments of the inter-annual variability, trend, and drivers of change. The comparative assessments of GHG emission time-series data provide information about sector- and country-specific dynamic performances towards fulfilling the commitment to the Kyoto Protocol. Therefore, the objective of this study was to detect the significance, direction, rate, and drivers of the observed trends of total CO₂-eq. GHG (GHG_{CO₂-e}) emission for the eight sectors of the 41 Annex-I countries over 23 years using data-driven models.

2. Material and methods

2.1. Data description

Countries that adopted the Protocol to the UNFCCC are required to report their GHG emission inventories (expressed in megatons, Mt) on an annual basis. The total and sector-specific annual GHG_{CO₂-e} (CO₂, CH₄, N₂O, SF₆, PFCs and HFCs) emission data (Mt CO₂-eq./year) for the period of 1990–2012 were used in this study for a total of 41 Annex I countries including Turkey and the following eight sectors: energy, energy industries, industrial processes,

land use, land use change and forestry (LULUCF), manufacturing industries and construction (MIC), transportation, wastes, and agriculture and were acquired from the UNFCCC [46,47] (2012). Population density (PD, people per km²), and gross domestic product (GDP, per capita of USD) data (inflation-adjusted 2010 prices) between 2001 and 2011 were obtained for 31 countries from the World Bank [53], and the USDA-ERS International Macroeconomic Data Set [50], respectively. Similarly, data about contribution of renewable energy to total energy production (RE, %) were obtained from OECD [34].

The principle of measurement, reporting and verification (MRV) was elaborated by a number of decisions by the Conference of the Parties for enhancing mitigation actions against climate change. According to the MRV principle, what are measured, reported and verified occur through biennial update reports (BURs), national communications, international consultation and analysis (ICA), and key assumptions and methodologies associated with the Revised 1996 IPCC Guidelines for National Greenhouse Gas Inventories, and the Greenhouse Gas Inventory Software for non-Annex I Parties (NAIIS) that the UNFCCC secretariat developed which incorporated all the elements prescribed by decision 17/CP.8. The MRV principle involves (1) inventory of GHG emissions by sources and removals by sinks; (2) enhancement of GHG emission reductions and removals by sinks relative to a baseline scenario; and (3) reporting of progress in achievement and implementation of climate change mitigation and adaptation as well as support received/needed in the form of finance, technology and/or capacity-building. Parties make their national GHG inventories following the Revised 1996 IPCC Guidelines for National Greenhouse Gas Inventories complemented by the IPCC Good Practice Guidance and Uncertainty Management in National Greenhouse Gas Inventories (2000) and the IPCC Good Practice Guidance for LULUCF (2003) and choosing one of the three tiers of the IPCC inventory methodologies according to their data availability: Tier 1 represents the minimum, or default method, while Tiers 2 and 3 involve more elaborate either source category-specific or technology-based methods.

2.2. Direction of trends

The Mann-Kendall (MK) test, a non-parametric statistical test, was used in this study to detect long-term trends in GHG_{CO₂-e} emissions between 1990 and 2012. The MK test is not sensitive to missing values, outliers, and data distribution type [21,30]. The MK “S” statistic can be calculated using Eq. (1) [39]:

$$S = \sum_{k=1}^{n-1} \sum_{j=k+1}^n \text{sgn}(x_j - x_k) \quad (1)$$

where x_j and x_k are annual data values, and $\text{sgn}(x_j - x_k)$ is an indicator function that takes a value of -1 , 0 and 1 according to the sign of $(x_j - x_k)$ as presented below in Eq. (2) [51]:

$$\text{sgn}(x_j - x_k) = \begin{cases} 1 & \text{if } x_j - x_k > 0 \\ 0 & \text{if } x_j - x_k = 0 \\ -1 & \text{if } x_j - x_k < 0 \end{cases} \quad (2)$$

The positive and negative signs of S represent the upward and downward direction of a trend, respectively. The variance of S takes into account tie (repeated) values in the data as presented in Eq. (3) [22]:

$$\text{VAR}(S) = \frac{1}{18} \left[n(n-1)(2n+5) - \sum_{p=1}^q t_p(t_p-1)(2t_p+5) \right] \quad (3)$$

where q is the number of tied groups, and t_p is the value in the p th group. The normal test statistic Z is determined using Eq. (4)

[43,44]:

$$Z = \begin{cases} \frac{S - 1}{[\text{VAR}(S)]^{1/2}} & \text{if } S > 0 \\ 0 & \text{if } S = 0 \\ \frac{S + 1}{[\text{VAR}(S)]^{1/2}} & \text{if } S < 0 \end{cases} \quad (4)$$

The positive and negative values of Z indicate the strength and direction of a monotonic increasing or decreasing trend over time, respectively, and were tested using a two-tailed test for the null hypothesis (H_0 : the presence of no significant trend), and the alternative hypothesis (H_A : the presence of a significantly increasing or decreasing trend). The null hypothesis was rejected in favor of H_A if the calculated absolute value of Z is greater than $Z_{1-\alpha/2}$, which can be found from the standard normal cumulative distribution tables at the following four significant levels: $p < 0.1^+$, $p < 0.05^*$, $p < 0.01^{**}$, and $p < 0.001^{***}$. Kendall [21] reported that the use of standard cumulative distribution table to decide the statistical significance of Z test is reliable only if the number of data points (n) is ≥ 10 which is the case with the present study. Otherwise, S should be calculated and statistical significance of the test should be judged accordingly as reported by Gilbert [11].

2.3. Rate of trends

If a significant trend is detected by the MK test, the rate of change in $\text{GHG}_{\text{CO}_2\text{-e}}$ emissions can be computed estimating Sen's slope (Q , Mt $\text{CO}_2\text{-eq./year}$), which was developed by Sen [41]. Similar to the MK, this is a non-parametric test that is not sensitive to the presence of outliers and missing values, and statistical distribution of data [11]. The Sen's slope (Mt $\text{CO}_2\text{-eq./year}$) was calculated taking the median of the slopes (Q_{ij}) of all individual data pairs $x_i(t_i)$ and $x_j(t_j)$, where $j > i$ and $Q_{ij} = \frac{(x_j - x_i)}{(t_j - t_i)}$, and therefore, is a non-parametric version of a linear regression equation ($y(t) = Q \cdot \Delta t + B$). As with Sauvage et al. [39], Q (slope) and B (intercept in Mt $\text{CO}_2\text{-eq.}$) values were used to calculate country-specific rates of change (P) expressed in %/year as follows:

$$P = \frac{Q}{B} \cdot 100 \quad (5)$$

Upper (UCL) and lower confidence levels (LCL) around Q were also calculated at 95% and 99% confidence levels. Detection of trends and their associated rate of change over the study period were determined when $n \geq 10$ using the software tool MAKESENS [38].

2.4. Data-driven modeling

Multiple non-linear regression (MNL) models were best-fitted to predict total $\text{GHG}_{\text{CO}_2\text{-e}}$ emissions as a function of population density, GDP per capita, RE, year 2001–2011, and country (31 countries). The addition of the predictors to the model was carried out using the terms through order of up to two, and their interactions through order of two. The selection of significant predictors for the best-fit MNL model was based on the stepwise selection method with alpha-to-enter and -remove values set to equal 0.01 in a non-hierarchical manner. The predictor “country” was used in the model as the categorical (dummy) explanatory variable, with Turkey left out as the reference level. The other continuous predictors were standardized using the method of subtracting the mean from the observation. This standardization method was selected not only to help to reduce multicollinearity, and thus, to improve the precision of the coefficient estimates as our model contained highly correlated predictors, higher-order

terms, and interaction terms but also to interpret each predictor coefficient, using the original measurement scale.

Variance inflation factor (VIF) for multicollinearity, Durbin–Watson (D–W) statistic for autocorrelation, adjusted coefficients of determination (R^2_{adj}) for the goodness of fit, and predictive power (R^2_{pred}) based on leave-one-out cross-validation, and standard error (SE) for precision were considered to assess the performance of the model. Multicollinearity among predictors was assumed to exit when $\text{VIF} \geq 10$, while autocorrelation was assumed to be positively and negatively correlated when $\text{D–W}=0$, and $\text{D–W}=4$, respectively, and not to exist when $\text{D–W}=2$. In addition, Pearson correlation matrix was performed to capture the strength and direction of the linear relationships among the variables. Cluster analysis was used to classify countries into groups as a function of similarity levels of country-specific Q and B observations over the 23-year period. All the data analyses were performed using Minitab 17.0 version (Minitab Inc., State College PA, USA).

3. Results and discussion

3.1. GHG emission trends of sectors

Total global $\text{GHG}_{\text{CO}_2\text{-e}}$ emission reached 49 ± 4.5 Gt in 2010 growing on average by 1.0 Gt/year (2.2%) between 2000 and 2010 and by 0.4 Gt/year (1.3%) between 1970 and 2000 [18]. The contribution of the 41 countries considered in this study to the total global $\text{GHG}_{\text{CO}_2\text{-e}}$ of 49 Gt in 2010 was about 55% (26.8 Gt). The total $\text{GHG}_{\text{CO}_2\text{-e}}$ is illustrated cumulatively and according to the eight sectors of the 41 countries in Fig. 1 and decreased by 10.5% from 29.2 Gt in 1990–26.2 Gt in 2012 with an overall Sen's slope of -613 Mt/year over the 23-year period (Fig. 2). Out of the eight sectors, the five sectors of LULUCF, manufacturing industries and construction, agriculture, waste, and industrial processes exhibited a decreasing trend, while the remaining sectors of energy, energy industries, and transportation showed an increasing trend in decreasing order of $\text{GHG}_{\text{CO}_2\text{-e}}$ emission, respectively (Fig. 3). The highest and lowest Sen's slopes were 1.2 Gt $\text{CO}_2\text{-eq./year}$ for the energy sector and -1.0 Gt $\text{CO}_2\text{-eq./year}$ for LULUCF, respectively. Out of the sectors, the maximum rise and decline in $\text{GHG}_{\text{CO}_2\text{-e}}$ emission occurred from the energy sector of the United States at a rate of 42.5 Gt/year ($p < 0.01$) and from LULUCF of the Russian Federation at a rate of -32.8 Gt/year ($p < 0.001$), respectively (Table 1).

Energy, energy industries, and transportation were the most significant GHG emitters among the 41 countries which can be attributed to their existing heavy reliance on fossil fuels. These sectors thus provide the greatest potential to reduce GHG emissions through the adoption of renewable energy technologies. On average, the total $\text{GHG}_{\text{CO}_2\text{-e}}$ emissions from the energy, energy industries and transportation sectors increased at a rate of 1201,

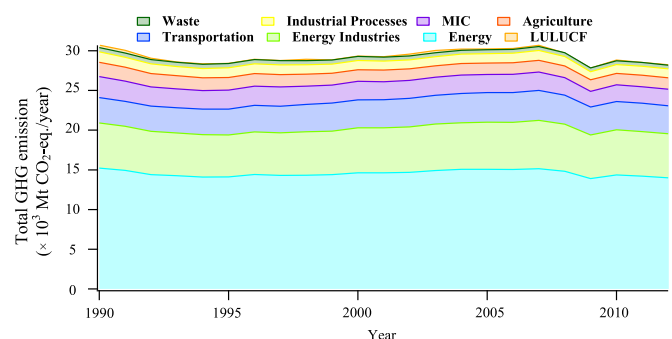


Fig. 1. Total CO_2 -equivalent GHG ($\text{GHG}_{\text{CO}_2\text{-e}}$) emission from eight sectors of 41 countries between 2001 and 2011.

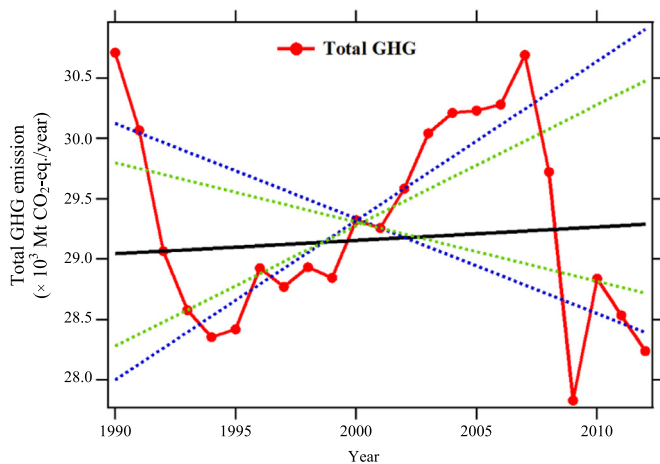


Fig. 2. Trend analysis of total CO₂-equivalent GHG (GHG_{CO₂-e}) emission from eight sectors of 41 countries between 2001 and 2011 (Sen's slope; 99% confidence level; and 95% confidence level).

858 and 822 Mt/year during the analyzed period for the 41 countries. The annual rate of sector-specific reductions in GHG_{CO₂-e} emissions on average ranged from −1021 Mt from LULUCF to −63 Mt from industrial processes (Table 1). Following LULUCF, manufacturing industries and construction (−330 Mt CO₂-eq./year), agriculture (−264 Mt CO₂-eq./year), and waste (−76 Mt CO₂-eq./year) were the sectors that achieved the greatest rates of reduction in GHG_{CO₂-e} emissions, respectively.

3.2. GHG emission trends of countries

Regardless of the significance of the trends detected, the largest emitters of the 41 countries over the study period were the United States for the sectors of energy, energy industries, transportation, agriculture, and industrial processes; Canada for LULUCF; and the Russian Federation for manufacturing industries and construction, and waste in decreasing order of GHG_{CO₂-e} emissions (Table 1). Similarly, the countries that cut their GHG_{CO₂-e} emissions at the highest rates over the study period were the Russian Federation for LULUCF, agriculture, and energy industries; Germany for energy, and transportation; the United States for manufacturing industries and construction, and waste; and Japan for industrial processes in decreasing order of GHG_{CO₂-e} emissions (Table 1). The three significantly highest rates of sector-specific GHG_{CO₂-e} emissions over the 23-year period belonged to the United States ($p < 0.01$), Turkey ($p < 0.001$), and Australia ($p < 0.001$) for energy; the United States, Japan, and Australia for energy industries ($p < 0.001$); Turkey ($p < 0.001$), Australia ($p < 0.001$), and Spain ($p < 0.05$) for industrial processes; Canada ($p < 0.05$), Ukraine ($p < 0.001$), and Germany ($p < 0.001$) for LULUCF; the Russian Federation ($p < 0.05$), Turkey ($p < 0.001$), and Canada ($p < 0.001$) for manufacturing industries and construction; the United States, Canada, and Spain for transportation ($p < 0.001$); the Russian Federation, Turkey, and Spain for wastes ($p < 0.001$); and the United States ($p < 0.05$), Canada ($p < 0.001$), and New Zealand ($p < 0.001$) for agriculture, respectively (Table 1). The three significantly highest reduction rates of sector-specific GHG_{CO₂-e} emissions occurred with Germany, Ukraine, and the United Kingdom for energy ($p < 0.001$); Poland ($p < 0.001$), Ukraine ($p < 0.01$), and Romania ($p < 0.001$) for energy industries; Japan, the United Kingdom, and Germany for industrial processes ($p < 0.001$); the Russian Federation ($p < 0.001$), the United States ($p < 0.05$), and Australia ($p < 0.05$) for LULUCF; the United States, Ukraine, and Germany for manufacturing industries and construction ($p < 0.001$); Germany for transportation ($p < 0.05$); the United States, the United Kingdom, and Germany for wastes

($p < 0.001$); and the Russian Federation, Ukraine, and the United Kingdom for agriculture ($p < 0.001$), respectively (Table 1).

Out of the 41 countries, a total of 18 countries (44%) achieved a significant rate of reduction in the total GHG_{CO₂-e} emission ($p < 0.05$), with the Russian Federation (−34.4 Gt/year) ($p < 0.01$), Germany (−10.9 Gt/year) ($p < 0.001$), and the United Kingdom (−8.3 Gt/year) ($p < 0.001$) as the highest achievers (Fig. 4). 12 countries (29%) significantly increased their rate of total GHG_{CO₂-e} emission ($p < 0.05$), with Canada (11.2 Gt/year) ($p < 0.01$), Turkey (10.0 Gt/year) ($p < 0.001$), and Spain (6.2 Gt/year) ($p < 0.001$) as the biggest contributors (Fig. 4). According to GHG_{CO₂-e} growth/reduction rates (P , %/year), the most reduction occurred with Lithuania, Romania, and Norway, whereas the most increase occurred with Turkey, New Zealand, and Cyprus (Fig. 4). Cluster analysis based on squared Euclidean distance and median linkage classified the 41 Annex-I countries into the five groups as a function of standardized P and Q data between 1990 and 2012, in decreasing order of similarity as shown in Fig. 5.

The addition of new data from developing and developed countries to our analysis is always possible as the number of (non-) Annex I countries has been updated continuously [48,49]. For example, the number of Annex I countries has risen to 43 in 2016 as far as the most recent data availability is concerned. However, the focus of the present study is on the detection of temporally monotonic trends as well as drivers and rates of change in total GHG_{CO₂-e} emissions.

3.3. Prediction and validation of total CO₂-equivalent GHG emissions

Ehrlich and Holdren [6] conceptualized the human-induced drivers of environmental change using the IPAT equation which stated that environmental impact (I) is the product of the following three factors: population (P), affluence (A), and technology (T). In the present study, as the indicators of P , A , and T , population density (PD), GDP per capita, and RE were operationalized, respectively. The best-fit MNLR model built in this study accounted for 99.8% of variation in the total GHG_{CO₂-e} emissions as a function of the following seven predictors: PD , year, country, the quadratic term of year, and the three interaction terms of GDP per capita by year, RE by year, and GDP^2 by year ($SE = 40$ Mt CO₂-eq./year; $p < 0.001$) (Table 2). Based on the leave-one-out cross-validation, the predictive power (R^2_{pred}) of the model was 99.8%, as well ($n = 340$). Except for PD , and country, the other predictors had a VIF value of < 10 which indicated a lack of multicollinearity issue, and also, the model had no autocorrelation issue according to its D-W statistic close to 2. There were 12 higher and 18 lower emitter countries than Turkey, with the United States, the Russian Federation, and Canada as the top three highest emitters and with the Netherlands, Belgium, and Luxemburg as the top three lowest emitters, respectively (Table 2).

Given the comparative magnitudes of their coefficients, the predictors of country, year, and PD were the most significant drivers of variation in the total GHG_{CO₂-e} emissions in decreasing order of significance. The total GHG_{CO₂-e} emission reduction rate was estimated at −8.34 Mt per year, while the total GHG_{CO₂-e} emission increase rate was estimated at 4.35 Mt/year given a one-unit increase in PD . As for the linear associations, the Pearson correlation matrix among the five variables of total GHG_{CO₂-e} emission, PD , GDP, RE , and year indicated that there were a total of the following four significant correlations between: GHG_{CO₂-e} and RE ($r = -0.215$; $p < 0.001$); PD and RE ($r = -0.431$; $p < 0.001$); GDP and RE ($r = 0.325$; $p < 0.001$); and GDP and year ($r = 0.126$; $p < 0.05$). These results suggest that the widespread adoption of sector-specific renewable energy technologies by the countries play a significant role in reductions in GHG_{CO₂-e} emission which in

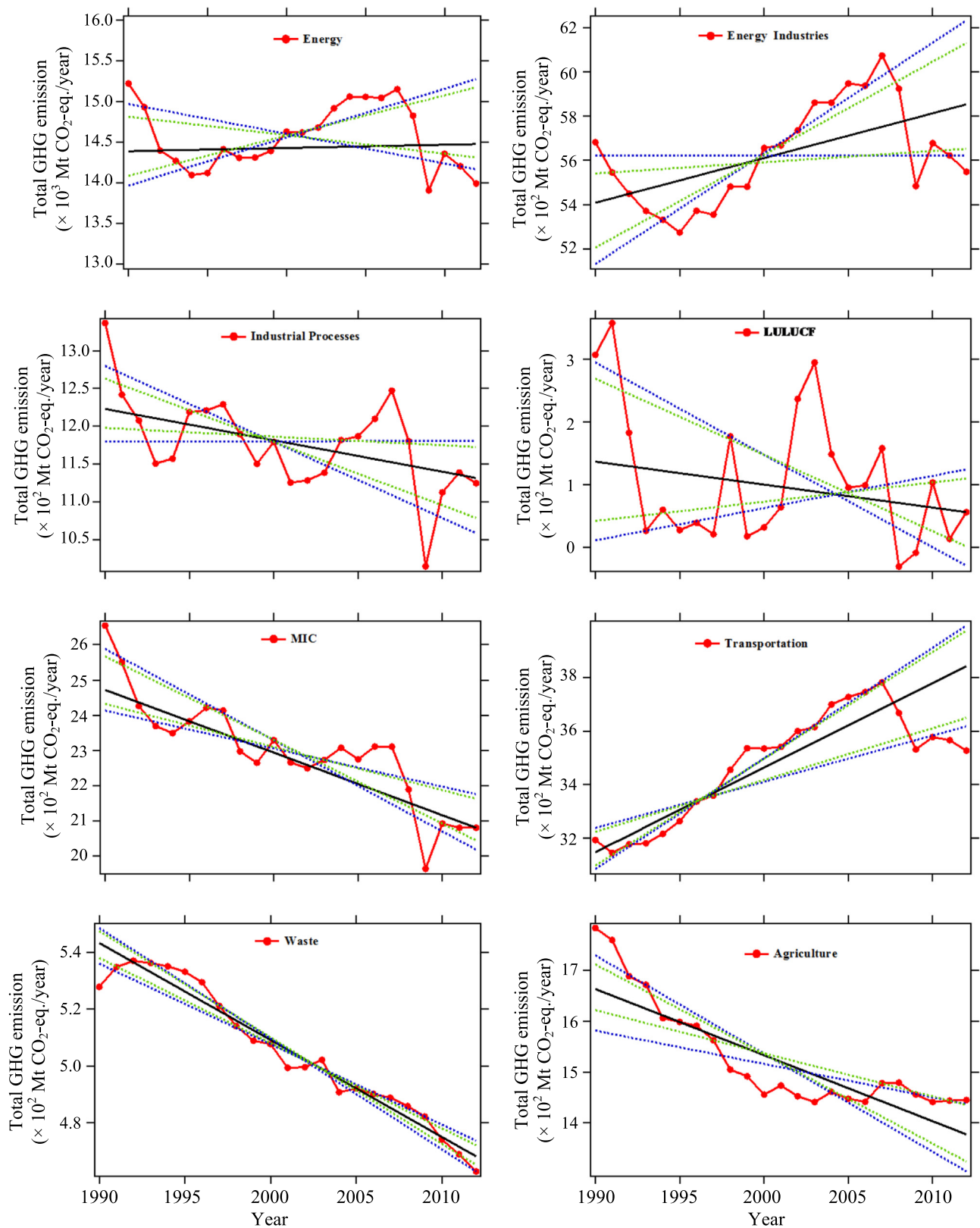


Fig. 3. Long-term trends of total CO₂-equivalent GHG (GHG_{CO₂-e}) emission according to the eight sectors of 41 countries between 2001 and 2011 (Sen's slope;99% confidence level; and 95% confidence level).

turn calls for the correction of market failures through carbon tax, renewable energy subsidies, publicly funded R&D, and governmental procurement programmes related to innovation and diffusion of renewable energy technologies.

4. Conclusions

The long-term trends and drivers of total GHG_{CO₂-e} emission (CO₂, CH₄, PFCs, HFCs, SF₆ and N₂O) from the eight sectors of the 41

Table 1

Direction (*Z*), rate (*Q*), and significance (*p*) of monotonic trends in the time-series of greenhouse gas emissions (Mt CO₂-eq./year) from eight sectors of 41 countries between 1990 and 2012 (*n*=23): LULUCF: land use, land use change and forestry; MIC: manufacturing industries and construction; *p* < 0.1⁺; *p* < 0.05⁺; *p* < 0.01⁺⁺; and *p* < 0.001⁺⁺⁺.

Country	Energy			Energy Industries			Industrial Processes			LULUCF			MIC			Transportation			Wastes			Agriculture			Total		
	<i>Z</i>	<i>Q</i>	<i>p</i>	<i>Z</i>	<i>Q</i>	<i>p</i>	<i>Z</i>	<i>Q</i>	<i>p</i>	<i>Z</i>	<i>Q</i>	<i>p</i>	<i>Z</i>	<i>Q</i>	<i>p</i>	<i>Z</i>	<i>Q</i>	<i>p</i>	<i>Z</i>	<i>Q</i>	<i>p</i>	<i>Z</i>	<i>Q</i>	<i>p</i>	<i>Z</i>	<i>Q</i>	<i>p</i>
Australia	6.39	6875.0	***	6.13	4580.6	***	5.02	488.9	***	−2.22	−3152.5	*	5.49	218.6	***	6.50	1395.3	***	−5.28	−225.2	***	0.16	38.8		2.27	4444.9	*
Austria	2.96	438.1	**	1.32	54.2	***	3.86	88.3	***	2.96	536.8	**	4.44	167.4	***	4.23	424.3	***	−6.44	−87.6	***	−4.60	−57.4	***	2.85	827.9	**
Belarus	−2.96	−726.0	**	−4.91	−556.8	***	3.91	113.3	***	0.79	73.1	***	2.75	91.3	**	−0.48	−48.3		5.81	225.5	***	−1.21	−159.1		−1.00	−168.4	
Belgium	−2.01	−532.2	*	−3.54	−195.0	***	−4.33	−236.7	***	−2.85	−28.5	**	−4.38	−428.6	***	5.18	330.1	***	−6.18	−88.9	***	−5.23	−117.5	***	−3.49	−1115.3	***
Bulgaria	−2.54	−486.1	*	1.11	121.5		−2.38	−159.3	*	4.91	241.2	***	−5.39	−582.9	***	4.91	224.9	***	−5.86	−81.4	***	−4.38	−198.7	***	−3.22	−788.8	**
Canada	3.75	6364.3	***	1.90	1136.0	+	−0.21	−16.5		2.01	4817.9	*	5.12	866.3	***	5.76	2685.6	***	4.23	126.2	***	3.86	409.0	***	2.64	11256.1	**
Croatia	3.17	309.6	**	1.69	57.8	+	2.75	50.7	**	4.23	100.0	***	−1.53	−15.1		5.44	186.6	***	6.55	25.1	***	−2.17	−17.3	*	3.49	460.5	***
Cyprus	5.60	174.5	***	5.81	109.8	***	2.64	11.4	**	2.69	4.05	**	0.32	1.39		6.02	55.0	***	5.70	14.4	***	0.63	1.25		5.44	216.8	***
Czech Republic	−4.54	−1126.4	***	0.69	90.2		−2.27	−90.7	*	1.95	118.5	+	−5.92	−853.9	***	5.39	629.7	***	6.29	43.5	***	−5.97	−167.7	***	−3.91	−1223.5	***
Denmark	−3.80	−752.9	***	−3.75	−582.0	***	−0.63	−23.8		−1.95	−178.5	+	−2.64	−53.4	**	4.17	121.6	***	−4.86	−24.8	***	−5.81	−141.1	***	−4.28	−1127.0	
Estonia	−2.32	−249.8	*	−3.01	−278.9	**	0.26	1.44		2.32	274.0	*	−2.22	−26.3	*	3.54	45.5	***	0.32	0.95		−2.91	−30.4	**	−0.26	−45.7	
Finland	0.48	58.8		1.95	308.7	+	3.96	104.5	***	−3.59	−576.7	***	−5.49	−138.9	***	3.43	72.1	***	−6.34	−108.0	***	−2.11	−7.93	*	−2.06	−592.1	*
France	−1.48	−1005.6		−1.16	−189.5		−5.76	−1070.1	***	−4.28	−823.4	***	−3.80	−704.9	***	2.17	533.4	*	−2.43	−83.3	*	−3.86	−333.3	***	−4.28	−3506.3	***
Germany	−5.97	−8637.6	***	−2.01	−1372.5	*	−3.75	−1207.1	***	4.01	1050.5	***	−4.49	−2078.4	***	−2.43	−1135.6	*	−6.55	−1432.7	***	−5.02	−469.2	***	−5.70	−10976.5	***
Greece	3.91	1526.7	***	3.96	797.1	***	−0.37	−18.7		−2.01	−21.7	*	−0.63	−45.7		4.75	434.3	***	−3.22	−59.7	**	−5.49	−90.5	***	3.33	1381.3	***
Hungary	−5.28	−697.5	***	−2.54	−240.2	*	−1.00	−51.1		−1.27	−79.4	***	−6.07	−386.4	***	4.65	315.3	***	3.49	25.9	***	−3.64	−82.7	***	−4.28	−904.9	***
Iceland	1.74	10.7	+	−2.91	−0.32	**	4.54	64.8	***	−6.60	−19.8	***	−0.85	−4.41		4.75	15.1	***	4.65	3.5	***	−2.27	−1.48	*	4.01	45.9	***
Ireland	3.80	685.1	***	1.11	86.0		−0.11	−6.7		−1.48	−64.0		2.96	79.3	**	4.81	494.2	***	−1.80	−20.5	+	−3.91	−139.0	***	1.95	489.1	+
Italy	1.11	1985.8		1.48	1183.7		−0.74	−136.7		−2.27	−663.7	*	−4.23	−860.6	***	3.01	962.0	**	−1.95	−153.9	+	−5.49	−366.9	***	0.32	392.2	
Japan	3.59	6369.1	***	5.12	5774.1	***	−4.54	−2319.4	***	−0.85	−301.7	***	−1.27	−846.0		−1.21	−893.9		−3.86	−463.4	***	−6.55	−248.1	***	1.43	1748.4	
Latvia	−3.80	−206.3	***	−5.81	−139.9	***	3.49	12.9	***	4.17	370.0	***	−5.07	−69.1	***	2.32	51.4	*	0.95	0.67		−1.06	−12.8		0.74	74.0	
Liechtenstein	0.53	0.42		4.17	0.06	***	6.66	0.45	***	5.55	0.13	***	−1.74	−0.21	+	−0.95	−0.16		4.91	0.02	***	1.64	0.05		1.21	1.03	
Lithuania	−3.17	−213.5	**	−5.18	−173.6	***	2.11	104.2	*	−2.64	−252.9	**	−2.80	−38.4	**	0.85	23.8		−1.06	−3.02		−0.74	−11.0		−2.96	−620.8	**
Luxembourg	0.79	40.5		4.44	60.3	***	−4.70	−31.3	***	−3.17	−11.9	**	−4.70	−148.1	***	5.18	193.7	***	−5.76	−1.20	***	−4.01	−4.38	***	−0.58	−38.7	
Malta	5.44	36.8	***	5.07	26.8	***	5.39	7.0	***	−5.42	−0.03	***	−1.95	−0.62	+	5.76	9.63	***	6.02	3.38	***	−1.90	−1.13	+	6.07	46.9	***
Netherlands	2.48	530.0	*	2.91	554.9	**	−4.44	−712.1	***	−0.90	−15.2	***	−4.75	−288.3	***	5.39	500.2	***	−6.60	−482.9	***	−6.29	−365.0	***	−3.80	−1180.2	***
New Zealand	4.49	520.9	***	2.64	129.7	**	5.12	82.6	***	3.80	404.4	***	0.26	4.60		6.02	274.0	***	3.01	24.9	**	4.12	238.6	***	4.81	1259.0	***
Norway	5.70	505.4	***	6.18	378.5	***	−4.17	−206.7	***	−5.07	−888.5	***	−2.32	−35.7	*	5.70	194.4	***	−6.44	−33.9	***	−4.75	−25.3	***	−4.44	−620.0	***
Poland	−3.27	−2682.9	**	−4.97	−2227.0	***	3.54	293.9	***	−1.90	−990.9	+	−3.86	−1075.6	***	5.86	1341.4	***	3.27	129.4	**	−4.07	−317.3	***	−3.49	−3763.5	***
Portugal	1.69	480.8	+	0.63	67.4		3.01	124.1	**	−4.01	−491.6	***	−1.48	−87.7		3.06	431.0	**	4.91	89.0	***	−3.17	−46.7	**	0.00	2.75	
Romania	−5.02	−2723.5	***	−4.86	−1564.7	***	−2.64	−282.9	**	2.38	112.9	*	−5.55	−1046.1	***	4.17	248.1	***	5.76	70.6	***	−4.33	−356.3	***	−4.38	−3138.9	***
Russian Federation	−0.05	−894.7		−1.21	−3343.9		0.85	882.6		−5.65	−32859.8	***	2.01	1857.2	*	0.16	450.3		4.91	1677.5	***	−5.39	−5694.2	***	−3.17	−34423.4	**
Slovakia	−5.92	−594.3	***	−4.07	−214.8	***	1.48	56.0		4.33	233.7	***	−5.76	−321.0	***	4.91	133.5	***	3.96	43.0	***	−5.92	−91.6	***	−3.54	−353.3	***
Slovenia	4.12	111.7	***	2.48	39.1	***	1.90	13.1		−0.79	−143.5	***	−4.01	−44.4	***	5.18	138.0	***	0.85	6.03	***	−2.75	−6.22	***	−1.37	−63.9	***
Spain	3.80	5945.4	***	2.85	1723.7	***	2.01	383.0	*	−5.55	−613.5	***	1.53	600.5		4.33	2583.2	***	6.29	274.3	***	1.06	82.5		3.59	6207.8	***
Sweden	−4.49	−516.4	***	−1.11	−51.0		1.90	24.5	+	0.63	136.1		−3.59	−204.8	***	3.27	95.6	**	−6.50	−90.0	***	−5.86	−64.1	***	−3.54	−558.5	***
Switzerland	0.48	17.3		5.44	79.7	***	4.07	59.2	***	2.17	71.1	*	−0.69	−6.48		4.70	97.5	***	−6.66	−22.2	***	−3.22	−22.0	**	1.06	72.6	
Turkey	6.18	8084.0	***	6.07	3978.6	***	5.86	927.8	***	−5.49	−654.6	***	3.54	1096.3	***	5.86	1165.4	***	5.86	906.4	***	−2.80	−233.9	**	6.39	10018.5	***
Ukraine	−5.02	−7936.6	***	−2.85	−2111.8	**	0.58	326.8		3.80	1968.3	***	−3.96	−2365.8	***	−0.85	−392.9		2.75	49.5	**	−5.07	−2500.0	***	−3.33	−8146.5	***
United Kingdom	−5.12	−4422.2	***	−1.80	−986.1	+	−4.49	−1290.5	***	−6.23	−502.3	***	−5.97	−1581.9	***	1.85	485.6	+	−5.07	−1484.2	***	−5.81	−701.4	***	−6.07	−8362.2	***
United States	3.01	42594.2	**	3.91	28049.1	***	1.11	1043.6		−1.90	−9049.8	+	−3.33	−4184.3	***	3.70	18830.8	***	−4.17	−1894.9	***	1.95	1485.6	+	1.53	17645.8	
Grand mean		1201			858			−63			−1021			−330			822			−76			−264			−613	

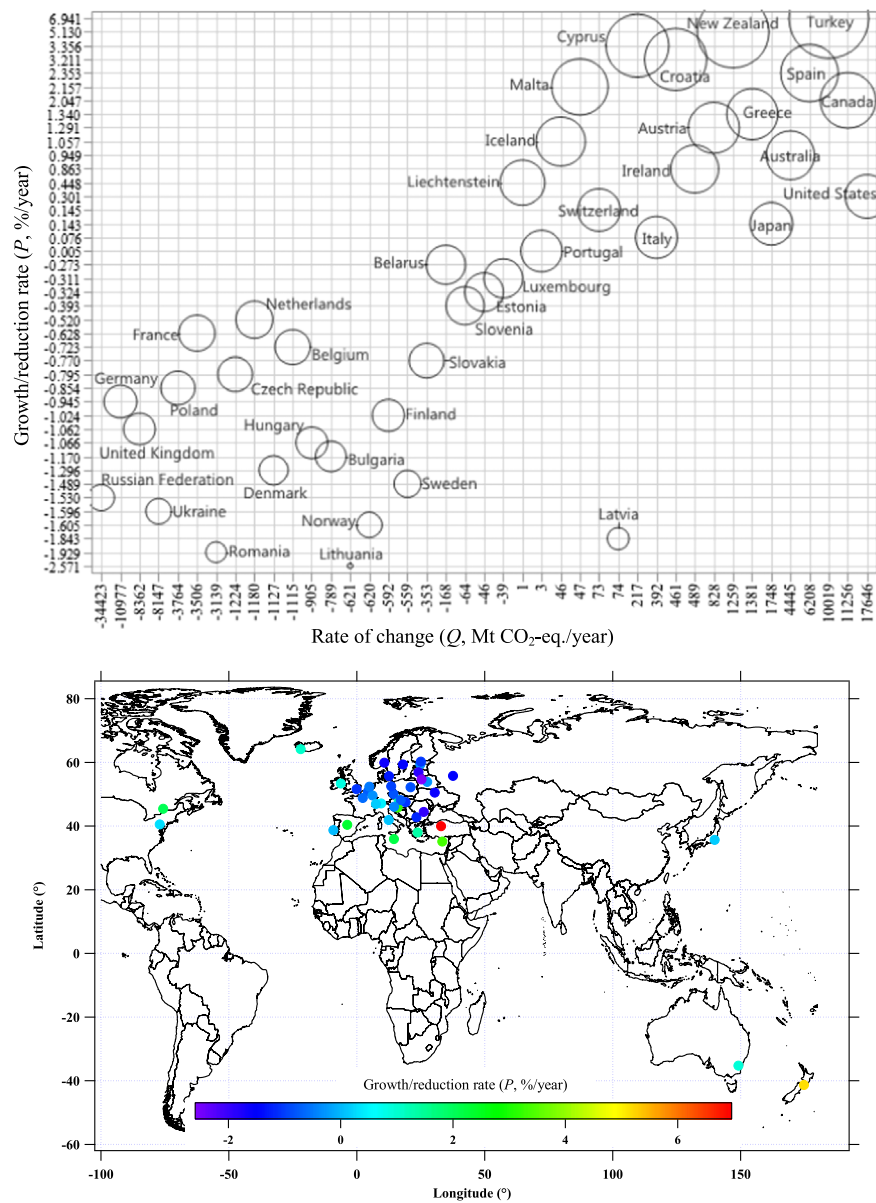


Fig. 4. Total CO₂-equivalent GHG (GHG_{CO₂-e}) emission between 1990 and 2012 from 41 Annex-I countries according to growth/reduction rates (P) (the greater the circle diameter is, the higher the growth rate is), Sen's slope (Q), and spatial distribution of P data.

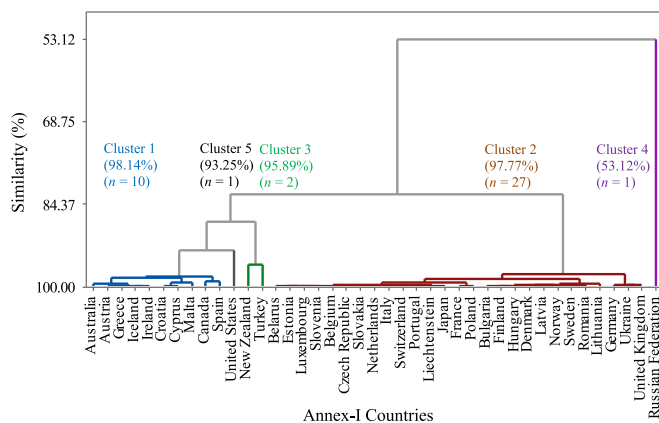


Fig. 5. Dendrogram for Annex-I countries as a function of GHG_{CO₂-e} growth/reduction rate (P, %/year) and Sen's slope (Q, Mt CO₂-eq./year) between 1990 and 2012.

Annex-I countries between 1990 and 2012 were quantified in this study. Results of the trend analysis indicate that the sectors of energy, energy industries, and transportation had the highest GHG_{CO₂-e} growth rates, whereas LULUCF, manufacturing industries and construction, agriculture, waste, and industrial processes had the highest GHG_{CO₂-e} reduction rates, respectively. Turkey, New Zealand, and Cyprus exhibited the highest GHG_{CO₂-e} growth rates (%/year), whereas Lithuania, Romania, and Norway showed the highest GHG_{CO₂-e} reduction rates (%/year), respectively. The highest and lowest Sen's slopes (Mt CO₂-eq./year) belonged to the United States and the Russian Federation, respectively. The global GHG_{CO₂-e} emission levels of 41–47 Gt/year by 2020 are reported to achieve the safe minimum standard (an increase in global average temperature by $\leq 2^\circ\text{C}$ above the pre-industrial levels by 2020) for avoiding the dangerous anthropogenic interference with the climate system under a wide range of assumptions and against the risks of long-term uncertainties as agreed by the UNFCCC Parties [37]. In the commitment periods of the Kyoto protocol between 2008 and 2012, and by 2020 during which the total GHG emission would be

Table 2
The best-fit multiple non-linear regression (MNLR) model of total CO₂-equivalent GHG (GHG_{CO₂-e}) emissions (Mt CO₂-eq./year) between 2001 and 2011, and its associated ANOVA results (SE = 40.0; R^2_{adj} = 99.8%; R^2_{pred} = 99.8%; D-W = 1.3; n = 340).

Term/source	Coefficient	SE	T	p	VIF	df	SS _{adj}	MS _{adj}	F	p
Intercept	446.1	37.4	11.94	0.001						
PD	4.35	1.08	4.04	0.001	3197	1	26,179	26,179	16.3	0.001
Year	−8.34	1.04	−7.99	0.001	2	1	102,506	102,506	63.8	0.001
Country [*]						30	403,245,670	13,441,522	8377.6	0.001
Australia	649.6	94.7	6.86	0.001	59					
Austria	−260.8	20.8	−12.53	0.001	3					
Belgium	−1293	281	−4.6	0.001	525					
Canada	852.2	93.6	9.11	0.001	58					
Czech Republic	−348.7	50.7	−6.87	0.001	17					
Denmark	−398.7	46.1	−8.65	0.001	14					
Estonia	−29.2	63.8	−0.46	0.647	27					
Finland	65.7	79.1	0.83	0.407	41					
France	91.7	33.6	2.72	0.007	7					
Germany	56	159	0.35	0.727	168					
Greece	−155.4	17.4	−8.94	0.001	2					
Hungary	−324.7	30.3	−10.72	0.001	6					
Iceland	82.4	94.1	0.88	0.382	59					
Ireland	−113.4	33.8	−3.36	0.001	7					
Italy	−252	119	−2.12	0.034	93					
Japan	−199	282	−0.71	0.481	526					
Luxembourg	−699	103	−6.81	0.001	70					
Netherlands	−1806	427	−4.23	0.001	1207					
New Zealand	74.3	80.8	0.92	0.359	43					
Norway	47.9	83.6	0.57	0.567	46					
Poland	−86.8	42.6	−2.04	0.043	12					
Portugal	−341.9	32.8	−10.42	0.001	7					
Russian Federation	1690.4	88.1	19.19	0.001	47					
Slovakia	−350.3	29.9	−11.7	0.001	6					
Slovenia	−325.9	20.8	−15.66	0.001	3					
Spain	67.3	17.1	3.94	0.001	2					
Sweden	24.9	73.9	0.34	0.737	36					
Switzerland	−675	108	−6.23	0.001	78					
United Kingdom	−345	176	−1.96	0.051	206					
United States	6033.2	63.2	95.54	0.001	26					
Year ²	−0.905	0.249	−3.64	0.001	1	1	21,274	21,274	13.2	0.001
GDP × Year	−0.000341	0.000052	−6.54	0.001	2	1	68,527	68,527	42.7	0.001
RE × Year	0.2667	0.0526	5.07	0.001	1	1	41,219	41,219	25.6	0.001
GDP ² × Year	0.000001	0.000001	4.78	0.001	2	1	36,712	36,712	22.8	0.001
Regression							406,206,150	11,283,504	7032.5	0.001
Error						303	486,151	1604		
Total						339	406,692,301			

^{*} Turkey was selected as the reference level for comparisons. PD: Population density (people per km²); GDP (per capita of USD); Gross domestic product; RE: renewable energy; SE: Standard error of coefficient; VIF: Variance inflation factor; df: Degrees of freedom; SS_{adj}: Adjusted sum of squares; MS_{adj}: Adjusted mean squares; D-W: Durbin-Watson statistic; and p: significance level.

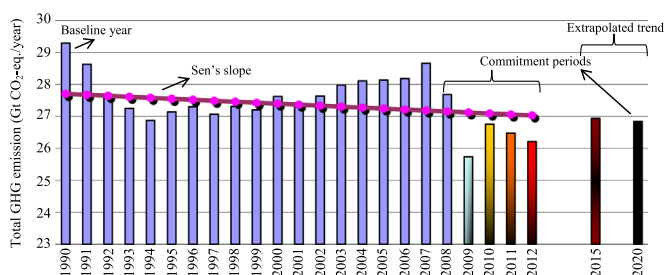


Fig. 6. Total CO₂-equivalent GHG emission trend ($y = 27705.61 - 30.574 \times \Delta t$) of 41 Annex-I countries between 1990 and 2012 and during the commitment periods.

reduced by at least 5% and 30% of the 1990 level, respectively, the 41 Annex-I countries released a total of about 26.2 Gt GHG_{CO₂-e} in 2012 (about 1.6 Gt CO₂-eq. below the initial commitment target of 27.8 Gt CO₂-eq.) and are projected to release a total of ca. 26.8 Gt CO₂-eq. in 2020 based on the current trend estimated (Fig. 6). This suggests that the total GHG_{CO₂-e} emission from the 41 Annex-I countries in 2020 is about 6.3 Gt above the 2020 target of 20.5 Gt. Considering that the 41 Annex-I countries used in this study did not include all

the other countries such as China and India and comprise ca. 55% of the total global GHG_{CO₂-e} emission (49 Gt) in 2010, the estimated downward trend needs to be further strengthened towards targeting the global GHG_{CO₂-e} emission levels of 41–47 Gt/year in 2020. For example, the total share of both China and India amounted to around 35% of the global GHG emissions in 2013 [5]. Total GHG_{CO₂-e} emissions of CO₂, CH₄, and N₂O in China were estimated at 7456.12 Mt in 2007 [3]. The validation of national GHG emission inventories against the real-time measurements is of great significance for the quantification of associated uncertainties and is considered a part of the IPCC Good Practice Guidance [23]. For example, Winiwarter and Rypdal [52] reported an uncertainty of 10.5–12% for the Austrian GHG emission inventory. Uncertainties of GHG emissions reported by countries vary significantly depending on their choice of estimation and assessment methods. Uncertainties associated with CO₂ emissions from fossil fuels consumption were estimated at 5% which was lowest when compared to the other source categories [31]. According to Lesiv et al. [28], mean relative uncertainty (%) as a proportion of total CO₂ emissions excluding LULUCF was estimated at 3.2 ± 2.5% for the 14 EU's Member States in 2008. Fine temporal and spatial resolution data for individual GHG concentrations play an important role in assessing uncertainties, short- and long-term

variations, and the effectiveness of mitigation measures towards the stabilization of atmospheric GHG concentrations.

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References

- [1] Arrhenius S. On the influence of carbonic acid in the air upon the temperature of the ground. *Philos Mag J Sci* 1896;5(41):237–76.
- [2] Bera PP, Francisco JS, Lee TJ. Identifying the molecular origin of global warming. *J Phys Chem A* 2009;113:12694–9.
- [3] Chen GQ, Zhang B. Greenhouse gas emissions in China 2007 inventory and input–output analysis. *Energy Policy* 2010;38(10):6180–93.
- [4] Dlugokencky EJ, Houweling S, Bruhwiler L, Masarie KA, Lang PM, Miller JB, Tans PP. Atmospheric methane level off: temporary pause or a new steady state? *Geophys Res Lett* 2003;30(19):5–14.
- [5] EDGAR. Trends in global CO₂ emissions: 2014 report. The Hague; 2014.
- [6] Ehrlich PR, Holdren JP. Impact of population growth. *Science* 1971;171:1212–7.
- [7] Erdogdu E. Turkish support to Kyoto Protocol: a reality or just an illusion. *Renew Sustain Energy Rev* 2010;14:1111–7.
- [8] Flegler JW, Carman KR, Weisenhorn PB, Sofranko H, Marshall T, Thistle D, Barry JP. Simulated sequestration of anthropogenic carbon dioxide at a deep-sea site: effects on nematode abundance and biovolume. *Deep Sea Res I* 2006;53:1135–47.
- [9] Forster PMD, Joshi M. The role of halocarbons in the climate change of the troposphere and stratosphere. *Clim Chang* 2005;71:249–66.
- [10] Fuglestedt JS, Bernsten TK, Godal O, Sausen R, Shine KP, Skodvin T. Metrics of climate change assessing radiative forcing and emission indices. *Clim Chang* 2003;58:267–331.
- [11] Gilbert RO. Statistical methods for environmental monitoring. New York: Pacific Northwest Laboratory; 1987.
- [12] Harnisch J, Borchers R, Fabian P, Maiss M. Tropospheric trends for CF₄ and CF₃CF₃ since 1982 derived from SF₆ dated stratospheric air. *Geophys Res Lett* 1996;23:1099–102.
- [13] Harnisch J, Eisenhauer A. Natural CF₄ and SF₆ on earth. *Geophys Res Lett* 1998;25:2401–4.
- [14] Haszpra L, Barcza Z, Hidy D, Szilagyi I, Dlugokencky E, Tans P. Trends and temporal variations of major greenhouse gases at a rural site in Europe. *Atmos Environ* 2008;42:8707–16.
- [15] Hoffert MI, Caldeira K, Benford G, Criswell DR, Green C, Herzog H, Jain AK, Khesghi HS, Lackner KS, Lewis JS, Lightfoot HD, Manheimer W, Mankins JC, Mauel ME, Perkins LJ, Schlesinger ME, Volk T, Wigley TML. Advanced technology paths to global climate stability: energy for a greenhouse planet. *Science* 2002;298:981–7.
- [16] IPCC. Climate change 2001: The scientific basis. Contribution of Working Group I to the Third Assessment Report of the Intergovernmental Panel on Climate Change. In: Houghton JT, Ding Y, Griggs DJ, Noguer M, van der Linden PJ, Dai X, Maskell K, Johnson CA (eds). New York: Cambridge University Press; 2001.
- [17] IPCC. Climate change 2007: The physical science basis. Contribution of Working Group I to the Fourth Assessment Report of the Intergovernmental Panel on Climate Change. In: Solomon S, DQin, M. Manning, Chen Z, Marquis M, Averyt KB, Tignor M, and Miller HL (eds). New York: Cambridge University Press; 2007.
- [18] IPCC. Summary for policymakers. In: Climate change 2014, mitigation of climate change. Contribution of Working Group III to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change. In: Edenhofer O, Pichs-Madruga R, Sokona Y, Farahani E, Kadner S, Seyboth K, Adler A, Baum I, Brunner S, Eickemeier P, Kriemann B, Savolainen J, Schlömer S, von Stechow C, Zwickel T, Minx JC (eds). New York: Cambridge University Press; 2014.
- [19] Kaygusuz K. Energy and environmental issues relating to greenhouse gas emissions for sustainable development in Turkey. *Renew Sustain Energy Rev* 2009;13:242–59.
- [20] Keeling CD. The concentration and isotopic abundances of atmospheric carbon dioxide in rural areas. *Geochim Cosmochim Acta* 1958;13(4):322–34.
- [21] Kendall MG. Rank correlation methods. London: Charles Griffin; 1980.
- [22] Kim KH, Shon ZH. Long-term changes in PM₁₀ levels in urban air in relation with the air quality control efforts. *Atmos Environ* 2011;45(19):3309–17.
- [23] Konstantinaviciute I, Bobinaite V. Comparative analysis of carbon dioxide emission factors for energy industries in European Union countries. *Renew Sust Energy Rev* 2015;51:603–12.
- [24] Kroeze C, Mozier A, Bouwman L. Closing the N₂O budget: a retrospective analysis. *Glob Biochem Cycles* 1999;13:1–8.
- [25] Kyoto Protocol. (http://unfccc.int/kyoto_protocol/items/2830.php); (accessed June 2011).
- [26] Le Quéré C, Moriarty R, Andrew RM, Peters GP, Ciais P, Friedlingstein P, Jones SD, Sitch S, Tans P, Arneeth A, Boden TA, Bopp L, Bozec Y, Canadell JG, Chini LP, Chevallier F, Cosca CE, Harris I, Hoppema M, Houghton RA, House JI, Jain AK, Johannessen T, Kato E, Keeling RF, Kitidis V, Klein Goldewijk K, Koven C, Landa CS, Landschützer P, Lenton A, Lima ID, Marland G, Mathis JT, Metzl N, Nojiri Y, Olsen A, Ono T, Peng S, Peters W, Pfeil B, Poulter B, Raupach MR, Regnier P, Rödenbeck C, Saito S, Salisbury JE, Schuster U, Schwinger J, Séférian R, Segsneider J, Steinhoff T, Stocker BD, Sutton AJ, Takahashi T, Tilbrook B, van der Werf GR, Viovy N, Wang Y-P, Wanninkhof R, Wiltshire A, Zeng N. Global carbon budget 2014. *Earth Syst. Sci. Data* 2015;7:47–85.
- [27] Lelieveld J, Crutzen P, Dentener FJ. Changing concentration, lifetime and climate forcing of atmospheric methane. *Tellus* 1998;50B:128–50.
- [28] Lesiv M, Bun A, Jonas M. Analysis of change in relative uncertainty in GHG emissions from stationary sources for the EU 15. *Clim Chang* 2014;124:505–18.
- [29] Maiss M, Brenninkmeijer CAM. Atmospheric SF₆: trends, sources and prospects. *Environ Sci Technol* 1998;32:3077–86.
- [30] Mann HB. Nonparametric test against trend. *Econometrica* 1945;13:245–59.
- [31] Marland G, Hamal K, Jonas M. How uncertain are estimates of CO₂ emissions? *J Ind Ecol* 2008;13:4–7.
- [32] Miller BR, Rigby M, Kuijpers LJM, Krummel PB, Steele LP, Leist M, Fraser PJ, McCulloch A, Harth C, Salameh P, Mühle J, Weiss RF, Prinn RG, Wang RHJ, O'Doherty S, Grealley BR, Simmonds PG. HFC-23 (CHF₃) emission trend response to HCFC-22 (CHClF₂) production and recent HFC-23 emission abatement measures. *Atmos Chem Phys* 2010;10:7875–90.
- [33] Mosier A, Kroeze C, Nevison O, Oenema S, Seitzinger S, van Cleemput O. Closing the global N₂O budget: nitrous oxide emissions through the agricultural nitrogen cycle-OECD/IPCC/IEA phase II development of IPCC guidelines for national greenhouse gas inventory methodology. *Nutr. Cycl Agroecosys* 1998;52:225–48.
- [34] OECD. Renewable energy (indicator). (<https://data.oecd.org/energy/renewable-energy.htm>); (accessed July 2014).
- [35] O'Neil BC. The jury is still out on global warming potentials. *Clim Chang* 2000;44:427–43.
- [36] Ramanathan V, Feng Y. Air pollution, greenhouse gases and climate change: global and regional perspectives. *Atmos Environ* 2009;43:37–50.
- [37] Rogelj J, McCollum DL, O'Neill BC, Riahi K. 2020 emissions levels required to limit warming to below 2 °C. *Nat Clim Chang* 2013;3:405–12.
- [38] Salmi T, Maatta A, Anttila P, Ruoho-Airola T, Amnell T. Detecting trends of annual values of atmospheric pollutants by the Mann-Kendall test and Sen's slope estimates-The excel template application MakeSens, RIVM, Bilthoven, The Netherlands; 2002.
- [39] Sauvage S, Plaisance H, Locoge N, Wroblewski A, Coddeville P, Galloo JC. Long term measurement and source apportionment of non-methane hydrocarbons in three French areas. *Atmos Environ* 2009;43:2430–41.
- [40] Seinfeld JH, Pandis SN. Atmospheric chemistry and physics: from air pollution to climate change. New York: John Wiley and Sons Inc; 1998.
- [41] Sen PK. Estimates of regression coefficient based on Kendall's tau. *J Am Stat Assoc* 1968;63:1379–89.
- [42] Shine KP, Bernsten TK, Fuglestedt JS, Skeie RB, Stuber N. Comparing the climate effect of emissions of short- and long-lived climate agents. *Philos Trans R Soc A* 2007;365:1903–14.
- [43] Shon ZH, Kim KH, Song SK. Long-term trend in NO₂ and NO_x levels and their emission ratio in relation to road traffic activities in East Asia. *Atmos Environ* 2011;45(18):3120–31.
- [44] Sicard P, Coddeville P, Galloo JC, Sauvage S. Trends in chemical composition of wet-only precipitation at rural French monitoring stations over the 1990–2003 period. *Water Air Soil Poll* 2007;7:49–58.
- [45] Turkish Ministry of Environment and Forestry. Climate change and performed studies, Ankara: Turkey (in Turkish); 2008.
- [46] UNFCCC. Greenhouse gas data-global map. Annex I, (<http://maps.unfccc.int/di/map/>); (accessed June 2011).
- [47] UNFCCC. Greenhouse gas profiles of Annex I parties, (http://unfccc.int/ghg_data/ghg_data_unfccc/ghg_profiles/); (accessed June 2011).
- [48] UNFCCC. Submitted biennial update reports (BURs) from non-Annex I parties, (http://unfccc.int/national_reports/nonannex_i_natcom/reporting_on_climate_change/items/8722.php); (accessed February 2016).
- [49] UNFCCC. 2016 Non-Annex 1 national communications, (http://unfccc.int/national_reports/non-annex_i_natcom/items/2979.php); (accessed February 2016).
- [50] USDA-ERS. International macroeconomic data set, (<http://www.ers.usda.gov/data-products/international-macroeconomic-data-set.aspx>); 2014.
- [51] van der Swaluw E, Asman WA, van Jaarsveld H, Hoogerbrugge R. Wet deposition of ammonium, nitrate, and sulfate in the Netherlands over the period 1992–2008. *Atmos Environ* 2011;45(23):3819–26.
- [52] Winiwarter W, Rypdal K. Assessing the uncertainty associated with national greenhouse gas emission inventories: a case study for Austria. *Atmos Environ* 2001;35:5425–40.
- [53] World Bank. Databank. (<http://data.worldbank.org/indicator>); 2014.